MATH 564 — Let's end well...

Remaining Topics:

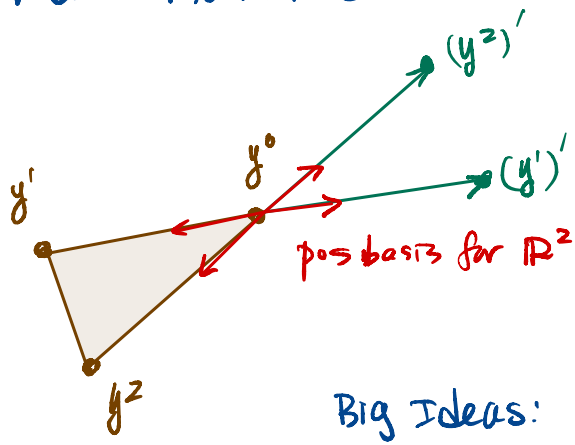
- MDSA (M)
- Project 5, Earth Resonances (M)
- Valve Design (W)
- Constrained DFO (F)
- Project 6, Comparative Study (F)
- Directional Direct Search (M)
- Valve Design Reveal (W)
- Sampling Strategies in $\Omega \subset \mathbb{R}^n$ (F)

Course Grades: Based on number of completed projects with some room for individual circumstances.
6 = A, 5 = A−, 4 = B+, 3 = B, 2 = B−, 1 = C+, 0 = C.

Alternative to Project 6: In-person competency review to be scheduled by student for the week of Dec 1.

Multi-Directional Search Algorithm (1990's)

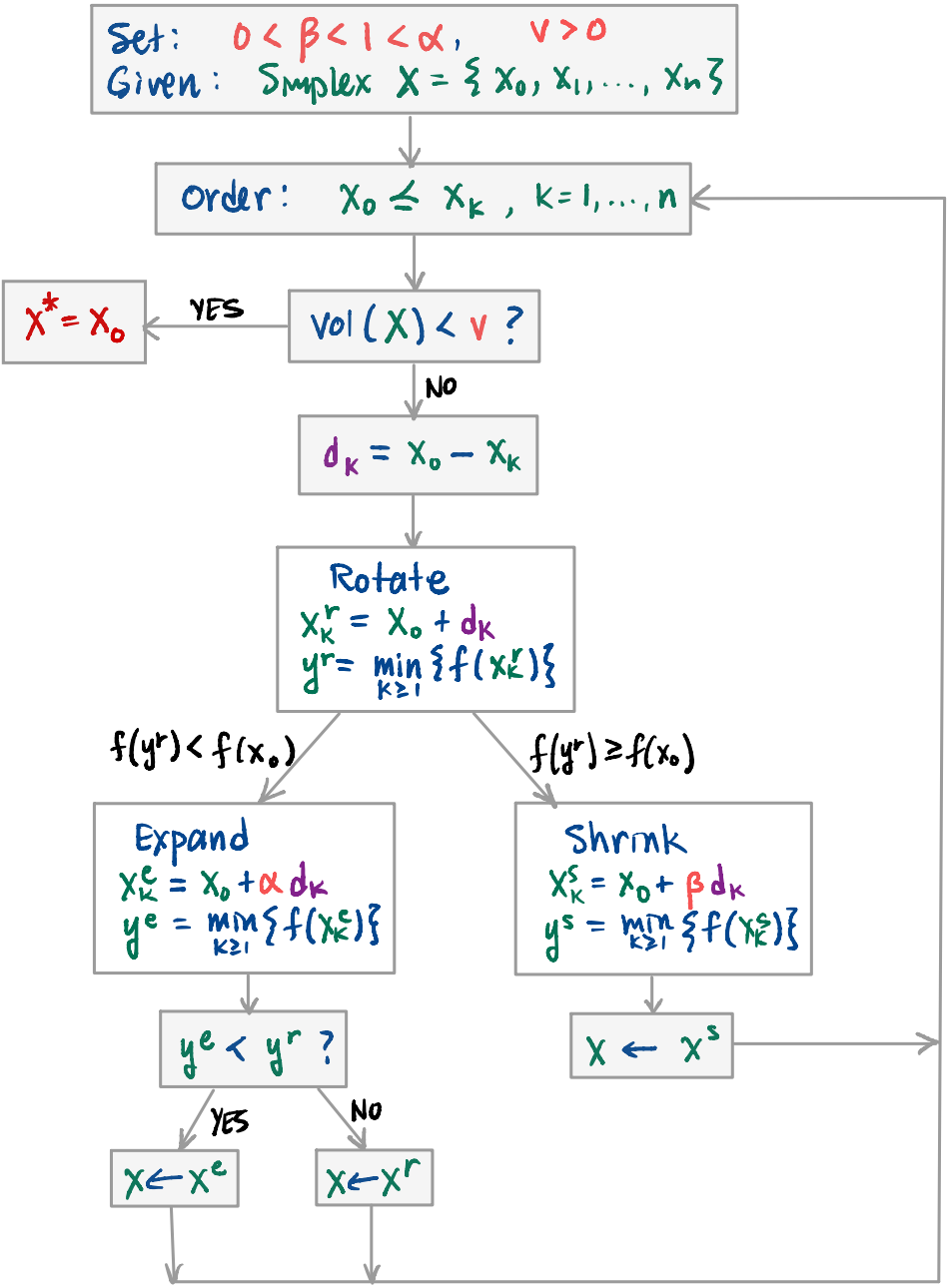
Nelder Mead Alternative



rotation about y^0

$(y^0 < y^1 < y^2)$

Big Ideas:

- only save the best point.
- search in a positive spanning set of directions.
- At least one of
  $\{y^k - y^0\} \cup \{(y^k)' - y^0\}$
  must be a descent direction.
- with small enough step size descent will occur.
- parallelizable !!
- search direction set is static so can fail for nonsmooth functions
- Try reflection, contraction, expansion

Multi-Directional Search Algorithm (MDSA)